Document 11 — Testing Strategy

Operator	[COMPANY NAME], trading as PlayStake
Owner	[ENGINEERING LEAD NAME], with sign-off by [COMPLIANCE LEAD NAME], Head of Compliance
Version	0.1 — draft for licence application
Review	Annually; on any new game, change to random number generation, or change to the RTS or testing requirements (Document 12)
Relates to	Remote gambling and software technical standards (RTS), and the Commission's testing strategy requirements for compliance with the RTS; licence condition on gambling software; Documents 12, 13

> Draft prepared from the platform as built on 30 September 2026. `[BRACKETS]` must be completed before submission. RTS paragraph numbers are for orientation and must be confirmed against the current RTS (Document 12). The split between third-party and in-house testing below must be confirmed with the Commission (§3.3).

1. Purpose

This strategy explains how PlayStake makes sure its gambling system meets the RTS before release and stays compliant through every change: what is tested, by whom, when, how it is recorded and who signs it off.

2. What we offer, and what needs testing

Product	Type	Random element	Outcome decided by
Tic-Tac-Toe	Peer-to-peer, pure skill	None	Server applies moves; first line of three wins
Darts 301	Peer-to-peer, skill with chance	Scatter of each dart around the aim point (server CSPRNG, Gaussian)	Server: board scoring, 301 rules, three rounds
Live stream matches	Peer-to-peer, skill, external game	None in PlayStake	Independent human referee, with dispute window

PlayStake does not offer house-banked games; players stake against each other and PlayStake takes a disclosed fee. All game logic and random number generation run on PlayStake's servers (Document 08 §3.1).

> **Note for the application:** Higher / Lower, a card game that was almost entirely chance, was withdrawn on 30 September 2026 so that everything offered fits a betting intermediary licence. Its

> past match records are kept; no new match can be started.

3. Approach

3.1 Third-party testing (approved test house)

We will engage a Commission-approved test house to test and certify, before real-money launch:

Item	What the test house checks	RTS
Random number generation	`src/lib/games/rng.ts`: Node.js `crypto.randomInt` (OS CSPRNG); unbiased integer draws; the uniform-to-Gaussian conversion (Box–Muller) used for dart scatter; no reuse or predictability; scaling free of modulo bias	RTS 7 (generation of random outcomes)
Darts rules and maths	Board geometry and scoring match the drawn board; scatter model; bust, checkout, three-round and draw rules match the published rules	RTS 3, 6, 7
Game records	Each move, draw and result is logged immutably and can be replayed	RTS 7

Test houses on the Commission's approved list include BMM Testlabs, eCOGRA, Gaming Associates Europe, Gaming Laboratories International (GLI), Global Lab, NFA / Eclipse, Quinel and RiskCherry (check the Commission's current list before engaging). **Selected test house: [TEST HOUSE]. Engagement date: [DATE].** The test house certificate and report are kept with this strategy and supplied to the Commission on request.

3.2 In-house testing

PlayStake tests everything else itself, under this strategy, with results recorded:

Area	How we test it	RTS
Customer information: rules, fees, payouts shown before commitment	Manual review against the game code at each release; screenshots kept	RTS 3
Safer gambling tools: deposit limits (gross, most restrictive applies), 24-hour delay on increases, first-deposit prompt, six-monthly review, breaks, reality checks, clock/elapsed time	Automated integration tests; manual check each release	RTS 12, 13, 14
Interrupted and abandoned games	Automated tests: void of unstarted matches, forfeit by the player to move, settlement of finished-but-unsettled matches	RTS 10
Collusion, cheating, bots	Controls reviewed each release; server-side outcomes proven by tests that reject client-declared results	RTS 11, 16

Area	How we test it	RTS
Ledger and escrow integrity	Automated tests of `transfer()`, escrow netting to zero, conservation; database triggers make entries append-only; daily production audit	Licence conditions on customer funds; RTS security requirements
Eligibility gate: KYC, GAMSTOP, breaks, account status	Automated tests	RTS 12, 14
Security controls	Document 13; independent penetration test; annual security audit	RTS security requirements

3.3 Points to confirm with the Commission

- Whether any RTS test requirement applies differently to peer-to-peer skill games where PlayStake is not a party to the bet.
- Whether a test house must also cover the platform functions listed in §3.2, or whether an in-house declaration is acceptable for them.

4. Automated testing on every change

- Every change goes through GitHub. CI runs lint, a type check, a dependency vulnerability audit (fails on critical findings), a malware scan and the full test suite against real PostgreSQL and Redis. A change can't be released unless CI passes.
- The suite (368 tests on 30 September 2026) includes, among others:
  - `tests/unit/ledger/` — double-entry transfers, escrow, audit invariants
  - `tests/unit/games/darts.test.ts` — board scoring matches the drawn board, bust, checkout, three-round result, draws
  - `tests/integration/game-integrity.test.ts` — only the server decides, declared results refused, event log is complete and append-only, darts scoring, sweep forfeits and settles correctly
  - `tests/integration/demo-hardening.test.ts` — no player can forge a session or settle someone else's bet
  - `tests/integration/compliance-gates.test.ts`, `limit-review.test.ts`, `session-idle.test.ts`, and the responsible-play, KYC, AML and complaints suites

5. Change control

Change class	Examples	Testing before release	Sign-off
Major	New game; any change to `rng.ts`, tic-tac-toe or darts rules, board geometry, the scatter model; change to fee or settlement arithmetic	Full automated suite; test-house re-test and new certificate for game/RNG changes; update rules shown to customers	Engineering lead + Head of Compliance; test-house certificate on file

Change class	Examples	Testing before release	Sign-off
Standard	Changes to safer-gambling tools, limits, eligibility, payments, ledger	Full automated suite plus new tests for the change; manual check of customer-facing text	Engineering lead + Head of Compliance
Minor	Copy, styling, non-gambling pages	Full automated suite	Engineering lead
Emergency	Security fix	Full automated suite; retrospective review within 5 working days	Engineering lead; Head of Compliance informed same day

Each release records: the commit, the CI run, the change class, any test-house reference, and sign-off. Game and RNG software versions are identified by the git commit of `src/lib/games/`.

6. Monitoring in production

- **Game fairness:** every draw is in `game_events`. A monthly statistical check (dart-scatter distribution against expectation) is to be run and recorded. [TO BUILD — outstanding item.]
- **Ledger:** the ledger-audit worker verifies every account, every transaction and system-wide conservation daily and alerts on any failure.
- **Results:** the anomaly worker flags skewed win rates, single-winner patterns, fast settlements and volume spikes.

7. Annual security audit

The RTS security requirements call for an independent annual security audit against the ISO/IEC 27001:2022 controls listed in Document 13. For a new licensee the first audit is due within **six months of the licence being granted**, then each year. The auditor must be independent and suitably qualified (for example ISO/IEC 27001 Lead Auditor, CISA, CISM or CISSP). The report goes to the Commission at securityaudit@gamblingcommission.gov.uk. Auditor: [AUDITOR]. First audit due: [GRANT DATE + 6 MONTHS].

8. Records

Test-house certificates and reports, CI results, release sign-offs, the monthly fairness check, and security audit reports are kept for at least five years.

Version	Date	Change
0.1	30 September 2026	First draft